Synchronization Markers for DNAFormer: Stage 4 Results

FYP, DNA data storage · 25 October 2026

Periodic markers clearly help under heavy deletion: at 8% they recover **52.4%** of strands vs **41.0%** for the end-block control, which has the same data and markers placed differently.

Version	Data bases	1%	3%	5%	8%	Errors at 8% (NED)	Data recovered at 8% (bases per 1,000 strands)
Marker-free	128	100%	99.8%	94.1%	38.8%	0.020	49,664
Periodic markers	96	100%	100%	96.1%	**52.4%**	**0.015**	**50,304**
End-block control	96	100%	100%	95.7%	41.0%	0.024	39,360

Exact strand recovery of fine-tuned DNAFormer by deletion rate; 1,000 unseen test strands per level; markers removed before scoring.

- **Spacing is what helps.** Periodic and end-block differ only in where the markers sit, so the 8% gain comes from synchronization.
- **Less data alone barely helps.** End-block is close to marker-free, and below 5% deletion all versions are near-perfect.
- **Worth it?** Yes as a finding, but on stored data the markers only break even (+1.3% at 8%). Limits: simulated data, one training run per arm.